EP2 - Exercise 4

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Setup and equations

We follow the collapse of a spherical top-hat overdensity in EdS and in flat ΛCDM, with radiation neglected at all epochs [1, 2], using $N = \ln a$, $A_H = 2 + d\ln H/dN$, $\vartheta = \nabla_x \cdot \mathbf{v}/(aH)$ and $g = 1/\delta$. The three formulations are implemented in separate files and integrated independently:

A: $\delta_{NN} + A_H\delta_N - \frac{4}{3}\frac{\delta_N^2}{1+\delta} - \frac{3}{2}\Omega_m\delta(1+\delta) = 0$ (as a first-order system in δ, δ_N),

B: $\delta_N = -(1+\delta)\vartheta, \qquad \vartheta_N = -A_H\vartheta - \frac{\vartheta^2}{3} - \frac{3}{2}\Omega_m\delta,$

C: $g_N = g(1+g)\vartheta, \qquad \vartheta_N = -A_H\vartheta - \frac{\vartheta^2}{3} - \frac{3\Omega_m}{2g}.$

After the change of variables the three right-hand sides agree to 10^{-14}. The linear δ_L is integrated in parallel in each system.

Initial conditions and shooting. At a_i we impose the linear growing mode, $\delta_{N,i} = f_i\delta_i$, $\vartheta_i = -f_i\delta_i/(1+\delta_i)$, $g_i = 1/\delta_i$, with f_i from the growth solver of Exercise 1. The terminal event is $\delta = \delta_\infty$ (A, B) or $g = \epsilon = 1/\delta_\infty$ (C), so the three stop at the same physical point and never cross the singularity. δ_i is tuned with a bracketed Brent root search on the collapse epoch, and $\delta_c = \delta_L(a_c)$. We use DOP853 with $\mathrm{rtol} = 10^{-10}$, $\mathrm{atol} = 10^{-2}\,\mathrm{rtol}\,a_i$, $a_i = 10^{-5}$ and $\delta_\infty = 10^8$. Brackets, event residuals ($\lesssim 10^{-14}$ in N) and settings are saved in the output tables.

Turnaround and virialization. The turnaround is the event $\vartheta = -3$ (reconstructed as $-\delta_N/(1+\delta)$ in A), with $\zeta_{\rm ta} = 1 + \delta(a_{\rm ta})$ ($\delta = 1/g$ in C). $y = R_{\rm vir}/R_{\rm ta}$ is the root of $4\lambda_{\rm ta}y^3 - 2(1+\lambda_{\rm ta})y + 1 = 0$ connected to 1/2 as $\lambda_{\rm ta} = \rho_\Lambda/\rho_{\rm sph,ta} \to 0$. In the formal-collapse convention, $\Delta_{{\rm vir},m} = \zeta_{\rm ta}(a_c/a_{\rm ta})^3 y^{-3}$ and $\Delta_{{\rm vir},c} = \Omega_m(a_c)\,\Delta_{{\rm vir},m}$.

EdS benchmarks

$\zeta_{\rm ta} = (3\pi/4)^2$ is exact; the small offsets elsewhere come from the finite threshold and the initial condition.

Cosmology grid

All quantities tend to EdS at high z_c, where Λ is negligible. At low z_c: (i) **growth** is slowed by Λ, so δ_c is slightly lower (1.6755 for $\Omega_{m0} = 0.3$, $z_c = 0$); (ii) the

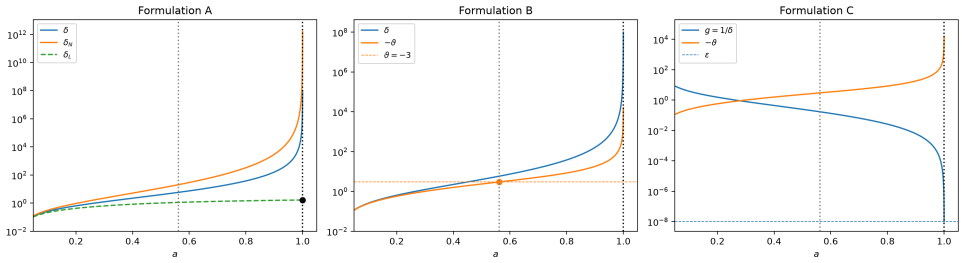


Figure 1: Flat ΛCDM, $\Omega_{m0} = 0.3$, $z_c = 0$. Dotted lines: turnaround (gray) and collapse (black). In A, δ_N diverges with δ; in B the turnaround is read at $\vartheta = -3$; in C, $g \to \epsilon$ but ϑ still diverges.

	δ_c	$a_{\rm ta}/a_c$	$\zeta_{\rm ta}$	$\Delta_{{\rm vir},m}$
target	1.68647	0.62996	5.5517	177.65
$z_c = 0$	1.68639	0.63000	5.5517	177.62
$z_c = 1$	1.68642	0.63000	5.5517	177.62
$z_c = 2$	1.68645	0.63000	5.5517	177.62

Table 1: EdS results; A, B and C agree to $\sim 10^{-10}$ in every entry. The threshold was not tuned to reproduce δ_c.

turnaround happens earlier relative to collapse and at higher $\zeta_{\rm ta}$ (6.84); (iii) the **virial prescription** gives $y < 1/2$ (0.4845), a more compact halo; (iv) for the **reference density**, $\Delta_{{\rm vir},m}$ grows well above $18\pi^2$ (340), while $\Delta_{{\rm vir},c} = \Omega_m(a_c)\Delta_{{\rm vir},m}$ falls below it (102).

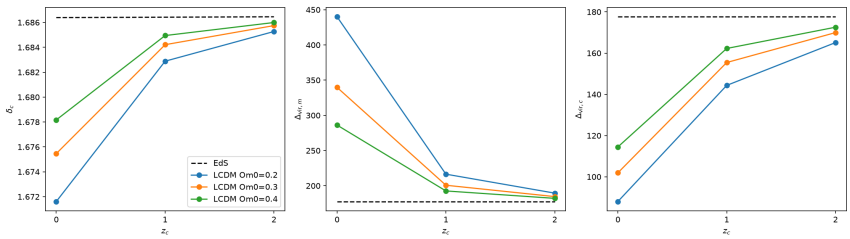


Figure 2: $\Omega_{m0} \in \{0.2, 0.3, 0.4\}$, $\Omega_{\Lambda 0} = 1 - \Omega_{m0}$, with the EdS baseline (dashed).

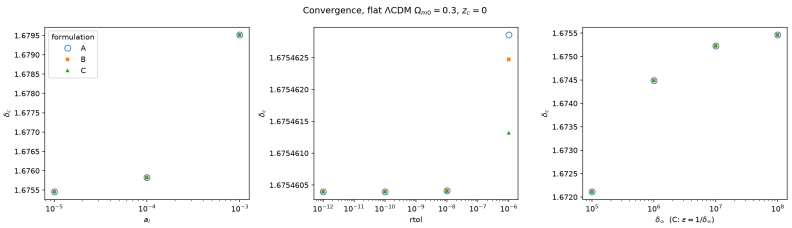


Figure 3: δ_c for $\Omega_{m0} = 0.3$, $z_c = 0$, changing one setting at a time (δ_i retuned in every run).

Setting changed	Value	δ_c	$a_{\rm ta}/a_c$	max diff. A, B, C
a_i	10^{-3}	1.67951	0.56146	6×10^{-11}
	10^{-4}	1.67583	0.56146	5×10^{-11}
rtol	10^{-6}	1.67546	0.56146	9×10^{-7}
	10^{-8}	1.67546	0.56146	6×10^{-9}
	10^{-12}	1.67546	0.56146	2×10^{-13}
δ_∞ ($\epsilon = 1/\delta_\infty$)	10^{5}	1.67212	0.56312	5×10^{-11}
	10^{6}	1.67449	0.56194	5×10^{-11}
	10^{7}	1.67523	0.56158	5×10^{-11}

Table 2: Same runs as Fig. 3. Fiducial ($a_i = 10^{-5}$, rtol $= 10^{-10}$, $\delta_\infty = 10^{8}$): $\delta_c = 1.67546$, $a_{\rm ta}/a_c = 0.56146$. Last column: largest relative difference in δ_c between A, B and C.

Convergence

The **tolerance** barely matters: the differences between A, B and C follow rtol. A finite **threshold** lowers δ_c by $\propto \delta_\infty^{-1/2}$ (linear growth missed before the true collapse), while an early **initial epoch** raises it by $\propto a_i$ (only the linear growing mode is imposed). These opposite errors can cancel by accident, so δ_∞ must not be tuned to δ_c; $a_{\rm ta}/a_c$ converges monotonically. **Failure modes.** A: $\delta_N^2/(1+\delta)$ blows up; B: ϑ diverges; C: g stays finite, but $1/g$ and ϑ diverge.

References. [1] Numerical Cosmology, Lecture 8 slides and notes, Sections.